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Osteopathic Considerations in the Pulmonary Patient

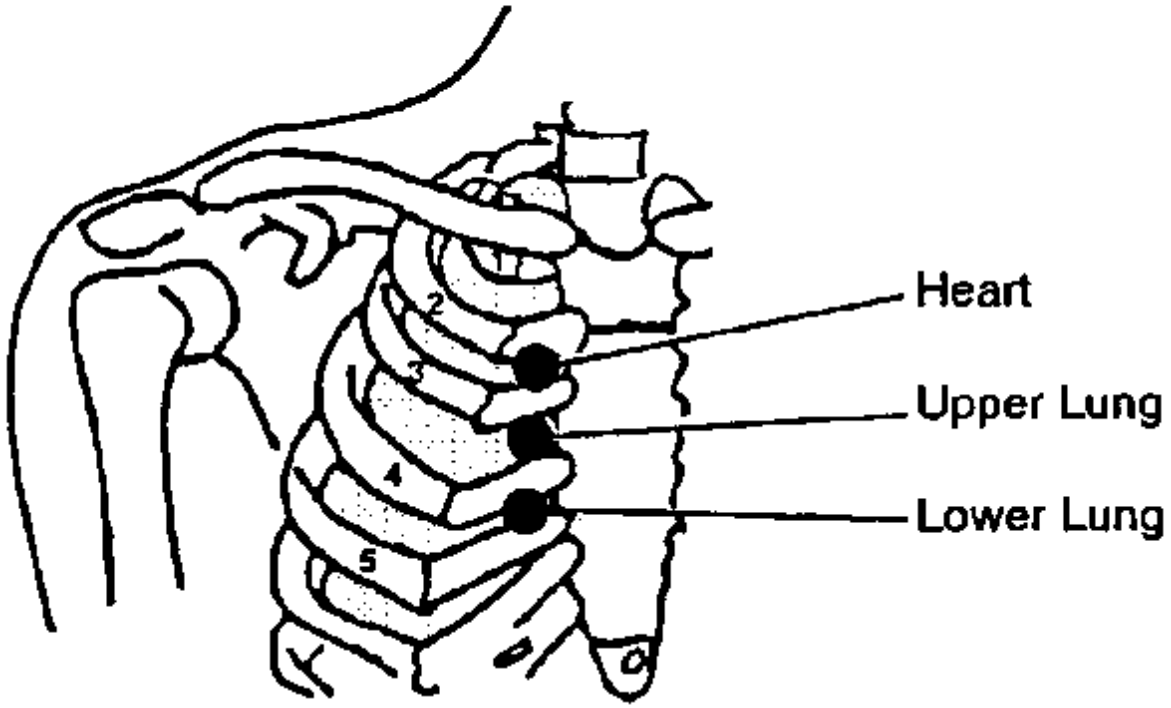
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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

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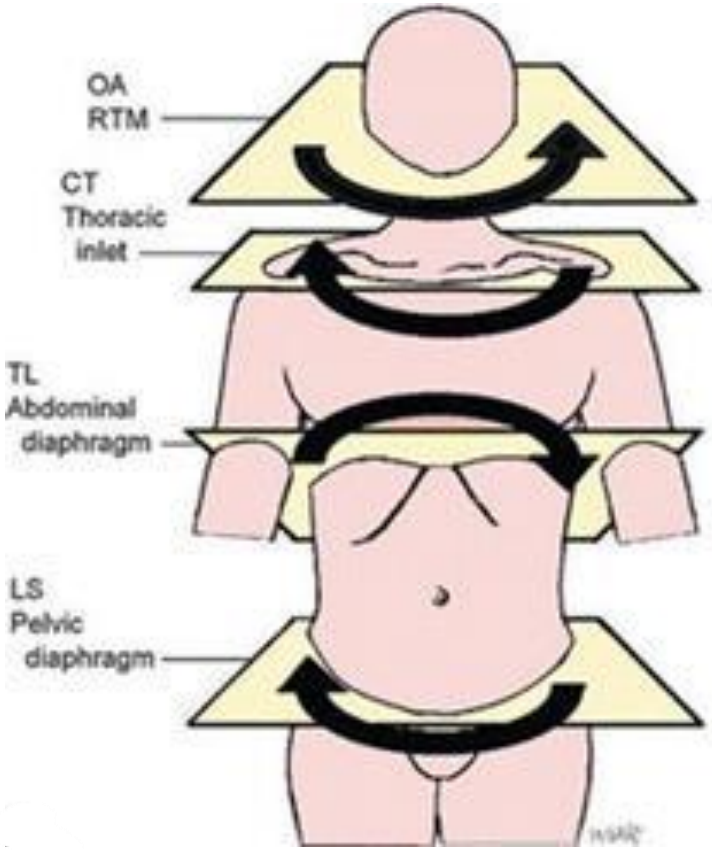
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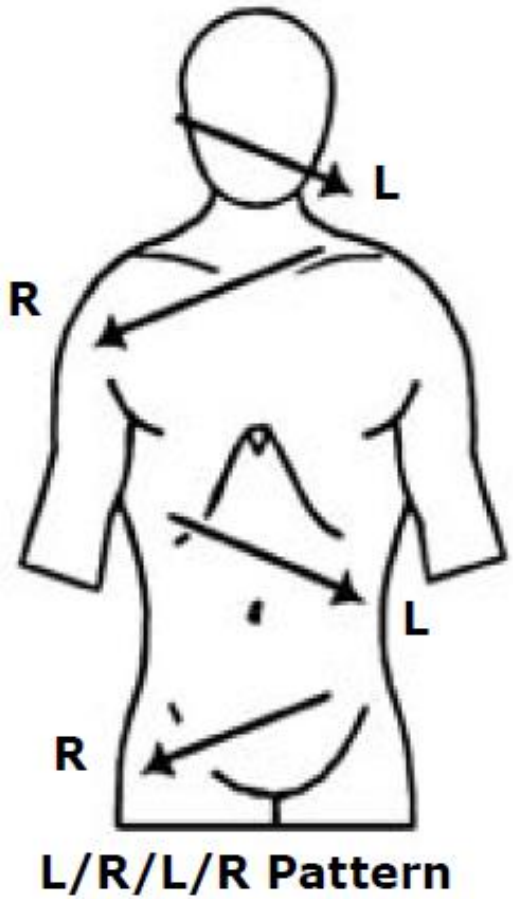
Cardiopulmonary System (T1-6)



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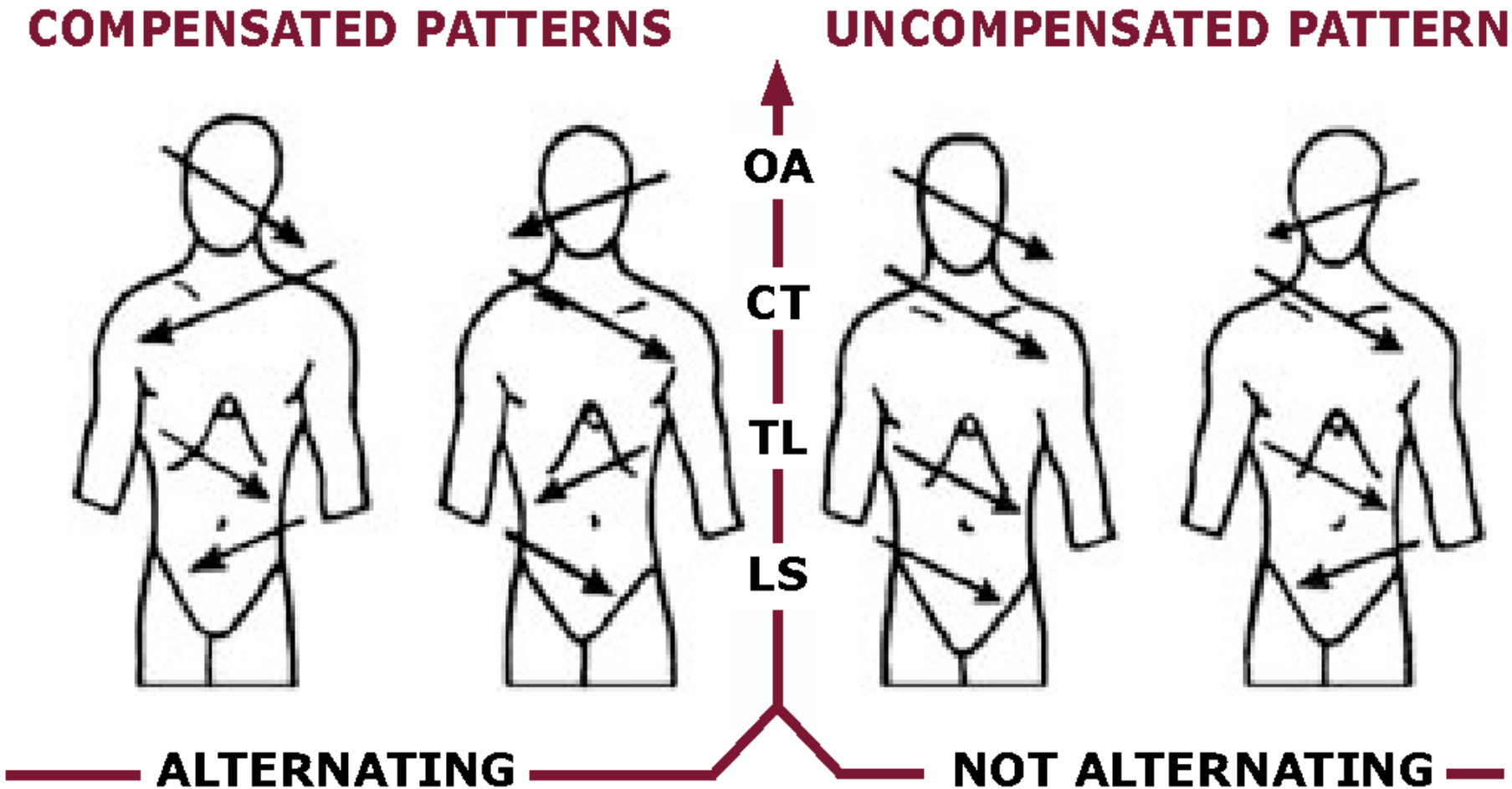




L/R/L/R Pattern

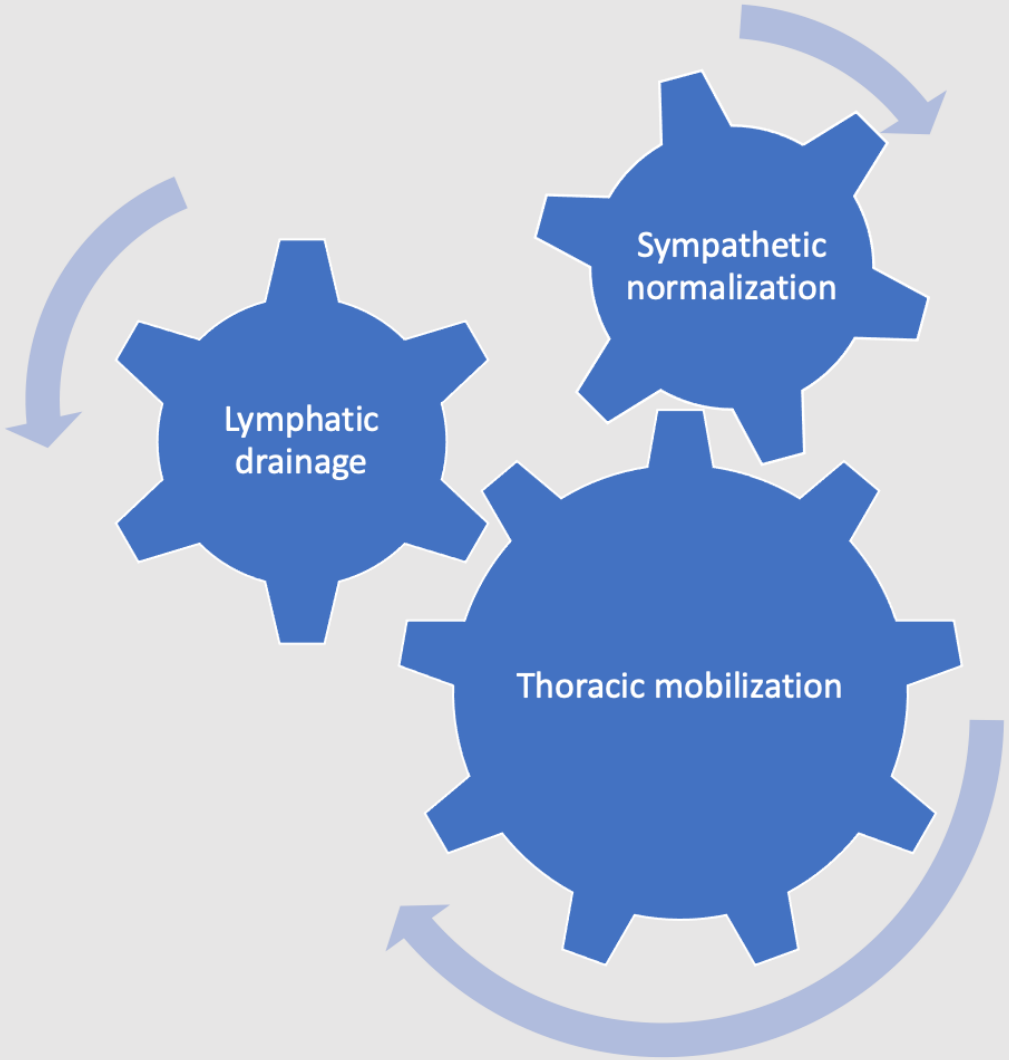
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Triad of treatment

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Three models reflect OMM in pulmonary disease:

- Respiratory/circulatory
- Biomechanical
- Neurological

ARDS

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- Biomechanical - early
  - Nociceptor activation – paraspinal muscle splinting → decreased cage compliance and thoracic vertebra somatic dysfunctions
  - Cough - rib somatic dysfunctions → decreased cage compliance and sympathetic chain ganglia dysfunction
- Biomechanical – late
  - Diaphragm fatigue (from decreased cage compliance and pulmonary edema)
    - Decreased pressure gradients for lymphatic flow
    - Ultimately ventilatory failure
- Diaphragm flattening – rapid, shallow, inefficient breathing

ARDS

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- Neurological facilitation
  - Nociceptor activation – efferent sympathetic discharge→ pulmonary vascular smooth muscle contraction
    - Decreased arterial supply
    - Decreased lymphatic clearance
  - Decreased amount of thinner, more productive mucous
    - Change to thicker consistency of mucous

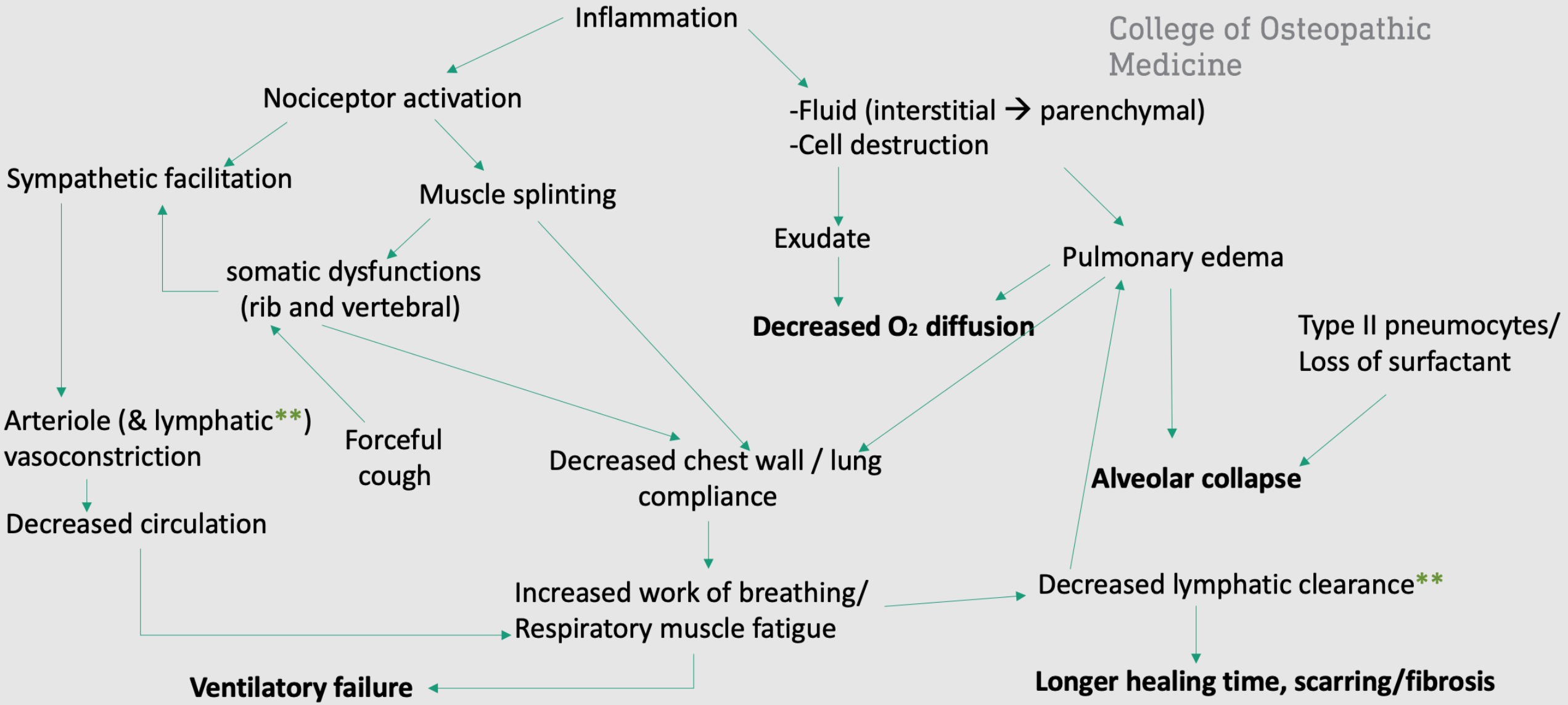
ARDS

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- Lymphatics
  - Inflammatory fluid accumulation
    - Pathogenic factors and inflammatory mediators linger
    - Decreased mobilization of APC to activate response
    - Ultimately protracted inflammatory response and delayed healing
  - Edema
    - Interstitial → alveolar: further barrier to oxygen diffusion
    - Weighs down lung parenchyma

ARDS

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OMM for the pulmonary patient

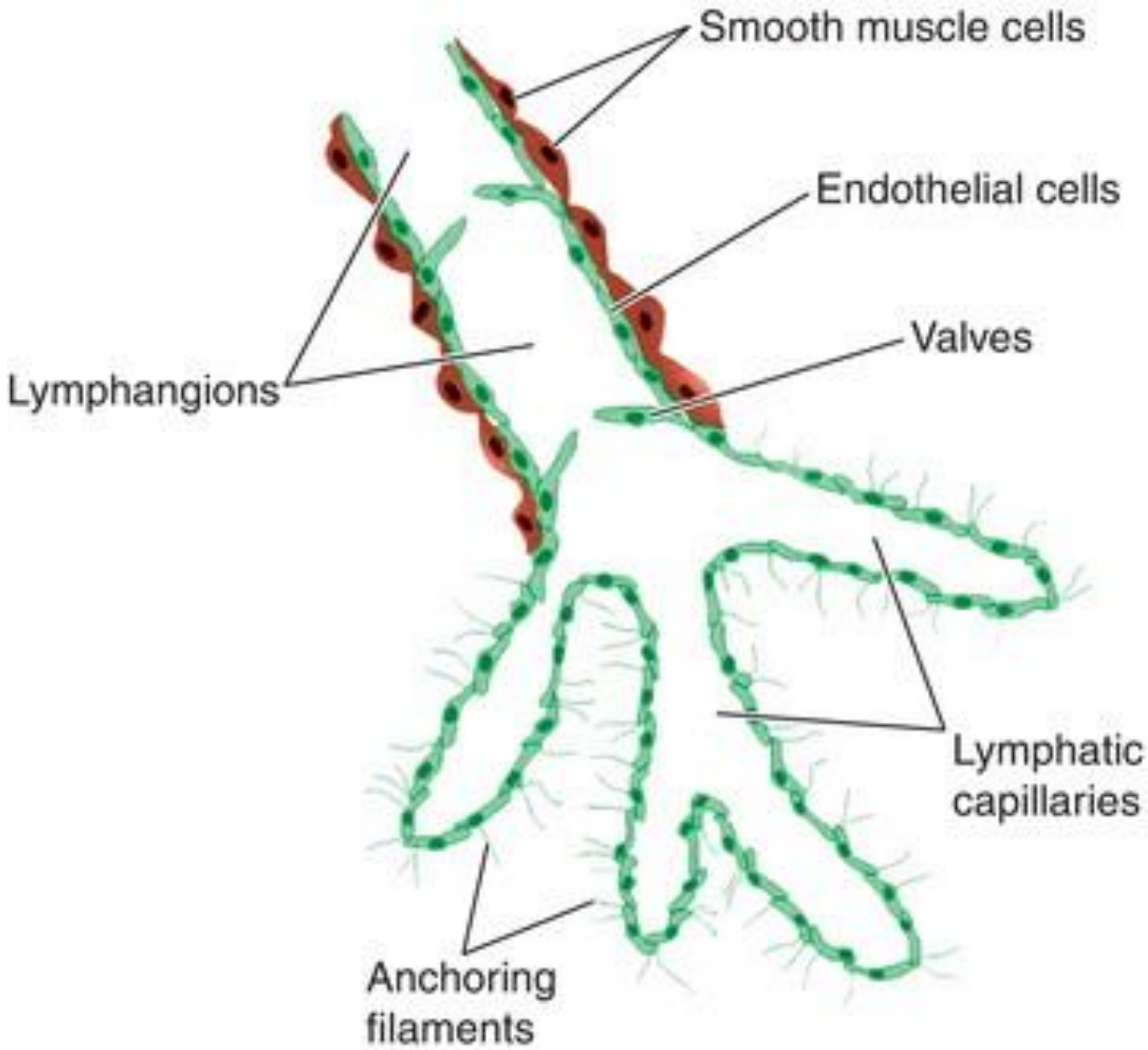
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- Core techniques
  - Rib raising
  - Thoracic inlet release
  - Respiratory diaphragm release
  - Vertebral somatic dysfunctions
    - Upper cervical (0-2): vagus
    - Mid cervical (3-5): phrenic
    - Lower cervical (6-7): adjacent to upper thoracic
    - Thoracic (1-6): mechanical and sympathetic
    - Thoracic (7-12): mechanical
  - Pedal pump

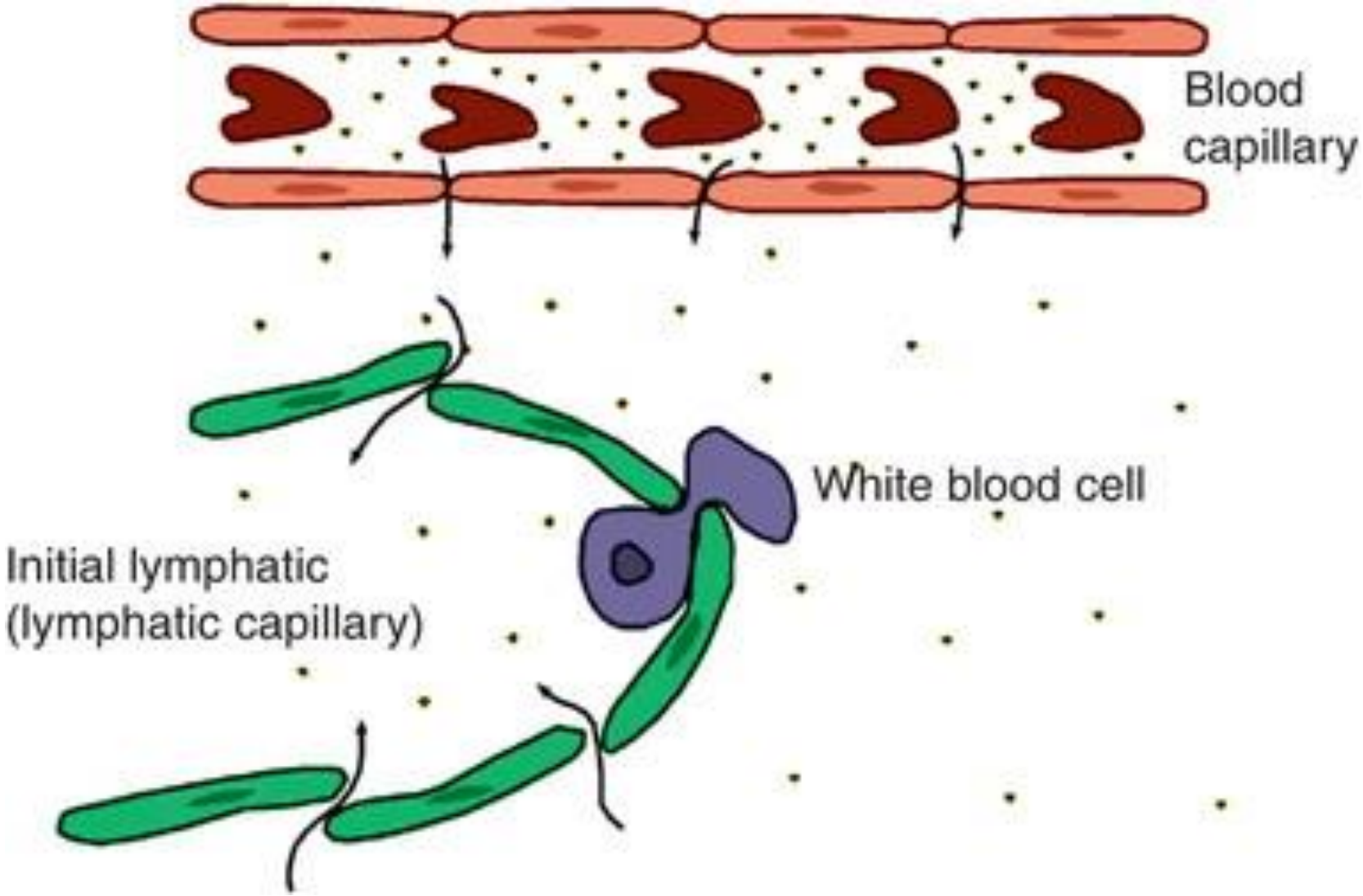
OMM for the pulmonary patient

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- Lymphatic drainage
  - Acuity may determine site of application
- Lymphatic opening channels
  - Thoracic inlet
  - Respiratory diaphragm
- Lymphatic pumps
  - Mobilizes formation and propulsion of lymph



Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018


Blood capillary
White blood cell
Initial lymphatic (lymphatic capillary)

References:

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